Multimodal LLMs

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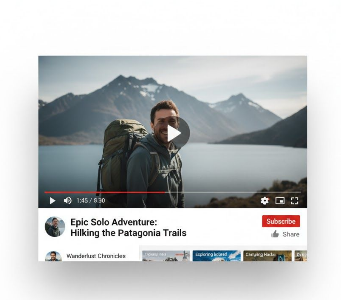
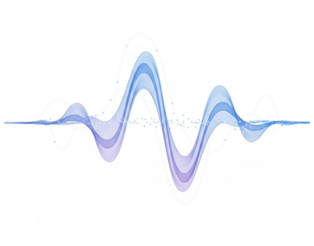
30.11.2025

What are multimodal models?

Multimodal models are needed when we want to combine information from different modalities like **images, audio and video**. These models are focused on adding **understanding** capabilities to accurately reason and answer.

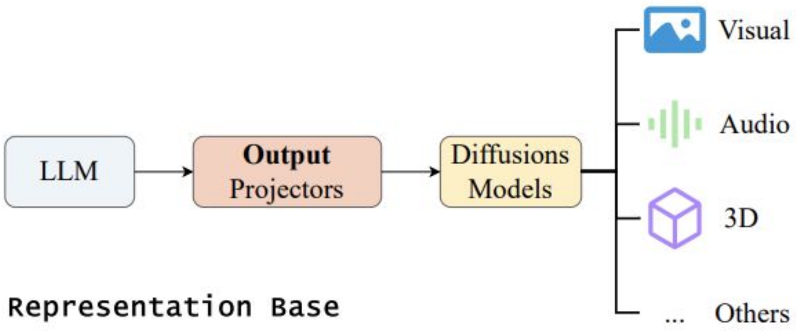
Multimodal models are typically constructed by extending a powerful pre-trained textual foundation model.





What is out of scope

Omni models are focused on **understanding and generation.** This is out of scope of this talk.



https://arxiv.org/pdf/2412.11694v3 From Specific-MLLMs to Omni-MLLMs

What is out of scope

Audio understanding would not be discussed in this lecture. Some pointers to popular encoders for audio:

- Wisper – encoder-decoder model from OpenAI 2022
- SoundStream – encoder-decoder with Residual Vector Quantization from 2021

https://arxiv.org/pdf/2509.17765 Qwen3-Omni

Introduction

LLMs operate on sequences of tokens and we want other modalities embedded in these sequences. Previously, there were approaches like Flamingo (2022)

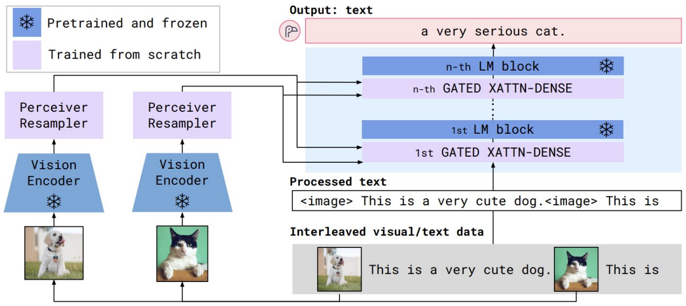


Figure 3: **Flamingo architecture overview.** Flamingo is a family of visual language models (VLMs) that take as input visual data interleaved with text and produce free-form text as output.

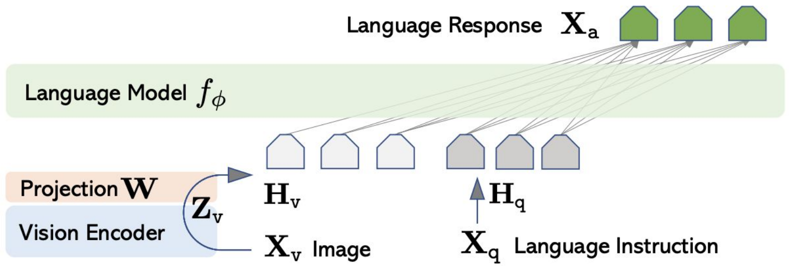


Figure 1: LLaVA network architecture.

https://arxiv.org/pdf/2204.14198

Plan

- **aligning modalities** – contrastive learning with language supervision
  - pre-training
- Visual Language Models
  - visual encoder
  - VL connectors
  - textual backbone
  - instruction tuning
  - evaluation
- Video Understanding
  - training-free adaptation of VLMs to video

MLLMs tasks – VQA

Goals:

- We want to have good question answering abilities given the visual/audio information.
- No regressions on text2text tasks like mathematics and coding.

I only had the sliced meat, how much do I need to pay? include a 18% tip

Here's the breakdown of the cost and the tip calculation, based on the receipt:

1. Identify the cost of the sliced meat:

- The receipt lists "1x Zürcher Geschnetzeltes + Rösti" at CHF 36.50. This is the sliced meat dish with a side of Rösti (fried potatoes).

2. Calculate the 18% tip:

- Tip amount = CHF 36.50 * 0.18 = CHF 6.57

3. Calculate the total amount to pay:

- Total = CHF 36.50 + CHF 6.57 = CHF 43.07

Therefore, you need to pay approximately CHF 43.07, including an 18% tip.

https://arxiv.org/pdf/2503.19786v1 Gemma 3

MLLMs tasks – agents

Main question: **how do we align two different modalities?**

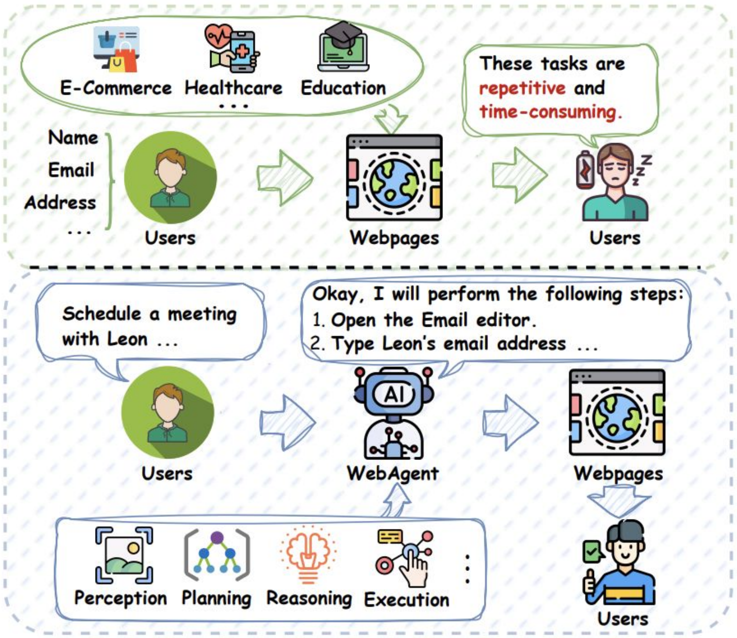


Figure 1: Illustration of basic web tasks and the pipeline of WebAgents. Given the user instruction, WebAgents autonomously complete tasks by perceiving the environment, reasoning action sequences, and executing interactions.

https://arxiv.org/pdf/2503.23350 – A Survey of WebAgents

How to train a general visual encoder?

We can train a ResNet/ViT for image classification but only limited labeled data is available.

A **caption** (textual description of an image) provides much more information than one class label.

The idea is to **scale** the captioning data and this way to improve the generalization of visual encoders.

https://arxiv.org/pdf/2103.00020 – Learning Transferable Visual Models From Natural Language Supervision

Image data with captions

MS-COCO (2015) contains 330k pairs and the captions are created by human annotators to describe the content of the images.

CLIP (2021) proposes 400M dataset collected from web WebImageText. LAION-5B was proposed in 2022.

Dataset	# English Img-Txt Pairs
Public Datasets	
MS-COCO	330K
CC3M	3M
Visual Genome	5.4M
WIT	5.5M
CC12M	12M
RedCaps	12M
YFCC100M	100M[2]
LAION-5B (Ours)	**2.3B**
Private Datasets	
CLIP WIT (OpenAI)	400M
ALIGN	1.8B
BASIC	6.6B

Table 1: **Dataset Size.** LAION-5B is more than 20 times larger than other public English image-text datasets. We extend the analysis from Desai et al. [14] and compare the sizes of public and private image-text datasets.

https://arxiv.org/pdf/2210.08402 LAION-5B

LAION-5B

This dataset:

- starts with Common Crawl
- images and alt-text tags from HTML
- automatic filtering based on similarity

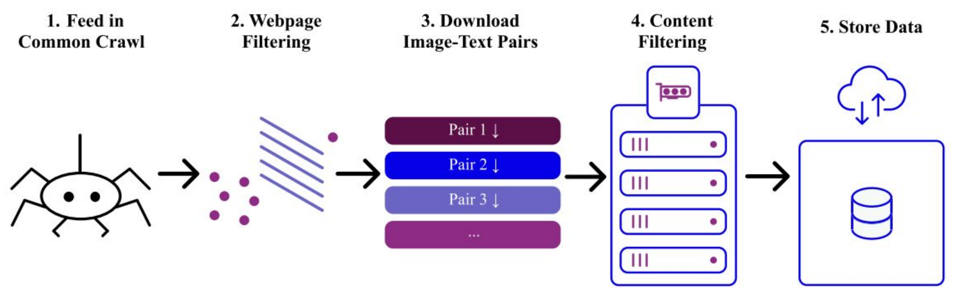


Figure 2: **Overview of the acquisition pipeline:** Files are downloaded, tracked, and undergo distributed inference to determine inclusion. Those above the specified CLIP threshold are saved.

https://arxiv.org/pdf/2210.08402 LAION-5B

Contrastive Language-Image Pre-training

Idea: learn image and text representations jointly in a shared embedding space.

- learn an image encoder f_image
- learn a text encoder f_text

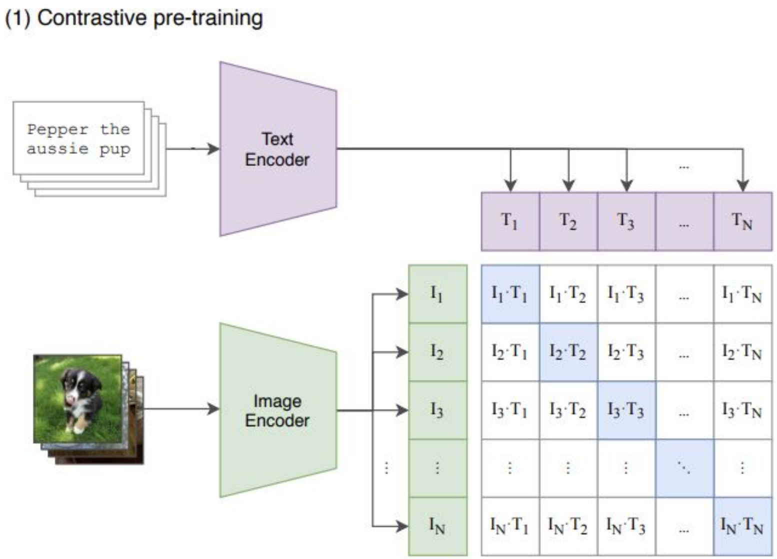
Representations for a paired image and text should be close to each other.

Representations for unpaired image and text should be far apart.

https://cmu-l3.github.io/anlp-spring2025/static_files/anlp-s2025-multimodal-i.pdf

Contrastive Language-Image Pre-training

Idea: given a batch of N pairs to train for classification.



https://cmu-l3.github.io/anlp-spring2025/static_files/anlp-s2025-multimodal-i.pdf

Contrastive Language-Image Pre-training

Idea: given a batch of N pairs to train for classification.

We average two cross-entropy losses for this task.

$$L((x_1, y_1), \ldots, (x_N, y_N)) =$$

$$-\frac{1}{2}\sum_{n=1}^{N}\left[\log \frac{\exp\left(f_I(x_n)^\top f_T(y_n)\right)}{\sum_j \exp\left(f_I(x_j)^\top f_T(y_n)\right)} + \log \frac{\exp\left(f_I(x_n)^\top f_T(y_n)\right)}{\sum_j \exp\left(f_I(x_n)^\top f_T(y_j)\right)}\right]$$

Softmax over images

Softmax over text

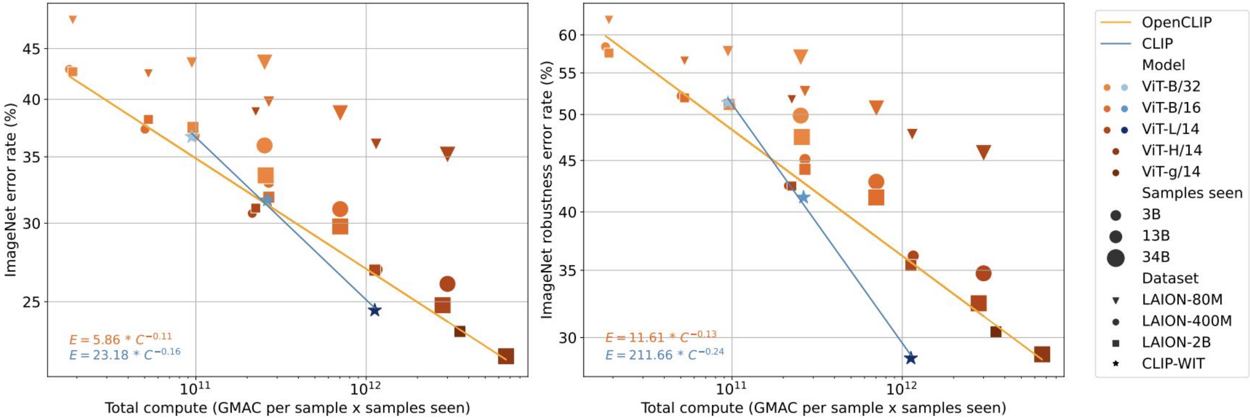
Contrastive Language-Image Pre-training

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

https://arxiv.org/pdf/2103.00020 CLIP

Pre-training and scaling

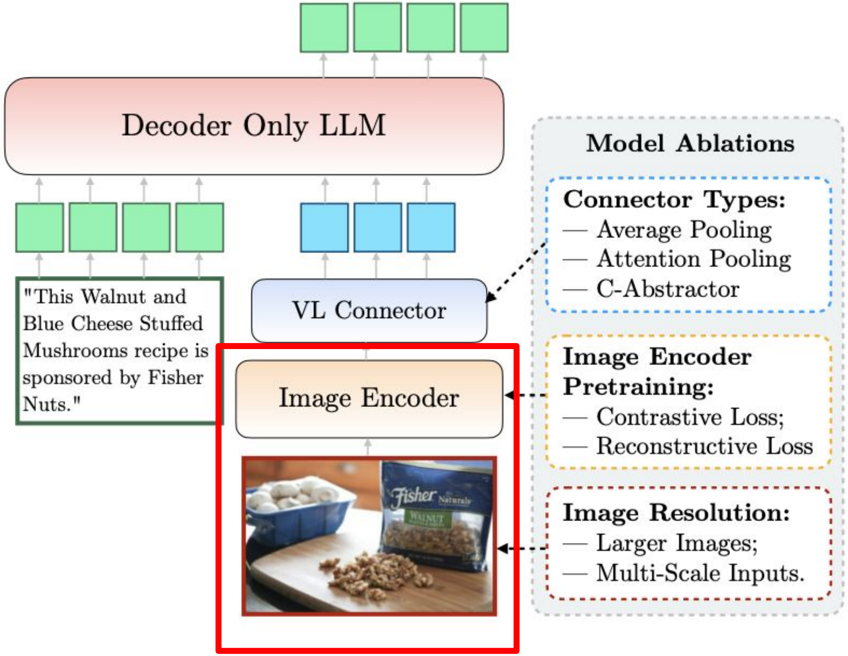
To train CLIP they used 32k batch size and 592 V100 GPUs.



(a) Relationship between total training compute and zero-shot classification performance on downstream tasks. **Left**: ImageNet performance. **Right**: average performance on five ImageNet robustness datasets (ImageNet-V2 [60], ImageNet-R [22], ImageNet-Sketch [74], ObjectNet [5], and ImageNet-A [24]). Scaling model size, data size, and samples seen leads to better performance on zero-shot classification. Models trained on OpenAI's WebImageText (WIT) show a stronger scaling than models trained on LAION.

https://arxiv.org/pdf/2103.00020 Reproducible scaling laws for contrastive language-image learning

CLIP in visual understanding



https://arxiv.org/pdf/2403.09611 MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training

PaLIGemma

Image tokens can also "lookahead" at the task at hand (prefix) in order to update their representation. This model is good for fine-tuning.

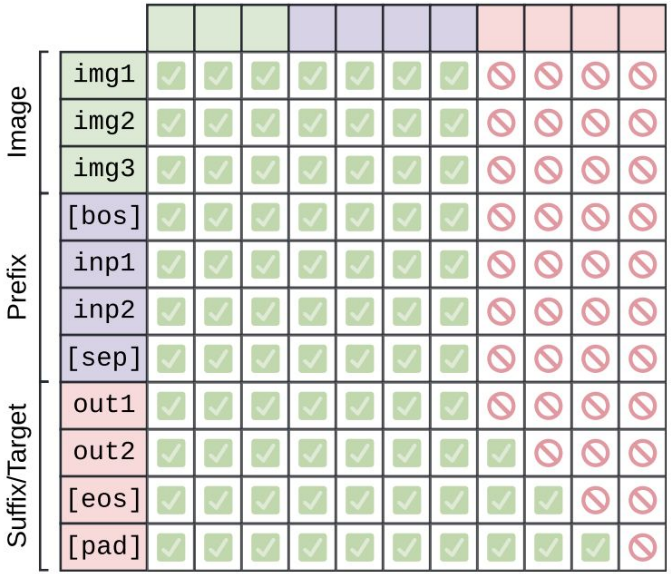


Figure 2 | PaliGemma's Prefix-LM masking: block attention throughout image and prefix, autoregressive attention on the suffix. Each square indicates whether the row can attend to the column.

https://arxiv.org/abs/2407.07726

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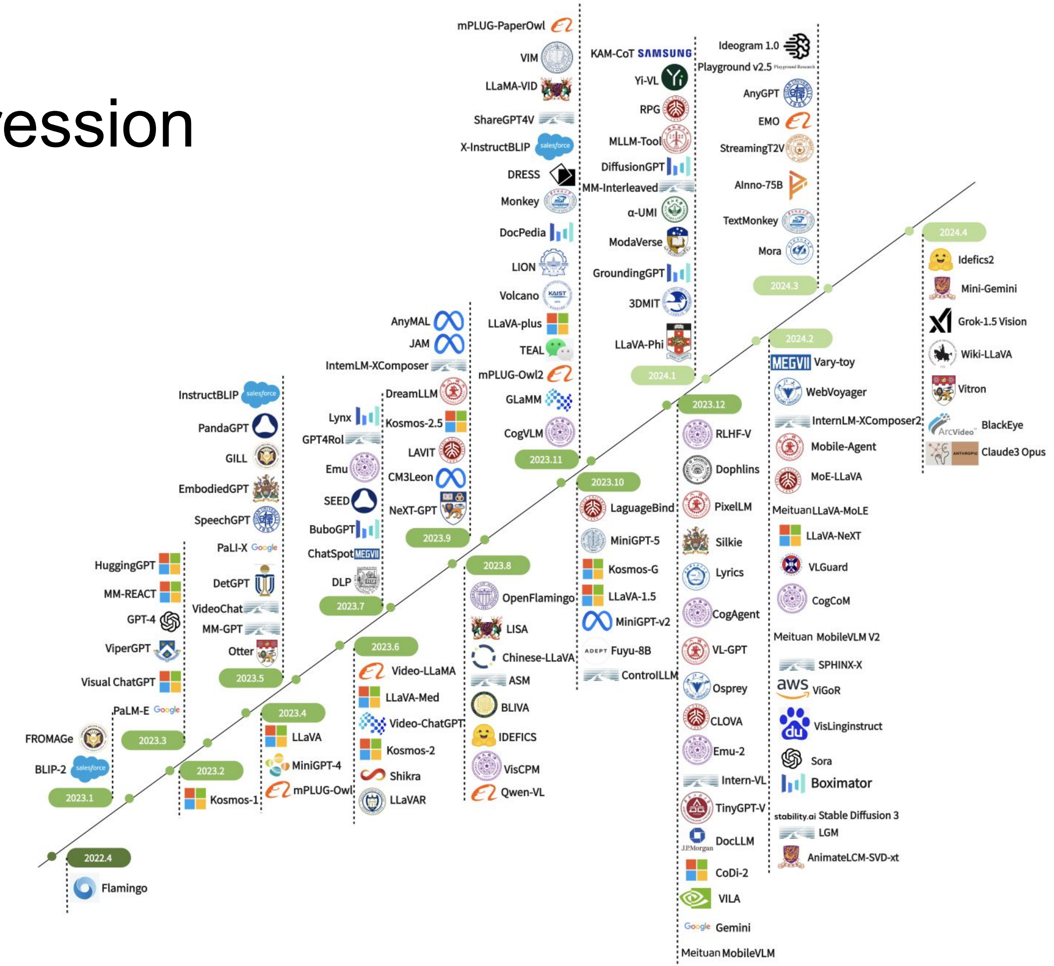
VLMs progression



https://arxiv.org/pdf/2408.01319 A Comprehensive Review of Multimodal Large Language Models: Performance and Challenges Across Different Tasks

Main models on arenas as of December 2025

licence	models
proprietary	Gemini (3, 2.5 pro), ChatGPT (4o, 4.5), Claude
OS	Qwen3-VL (Nov 2025), Mistral, Gemma 3

https://lmarena.ai/leaderboard/vision

Visual encoder

Contrastive training shows better performance than self-supervised methods.

The biggest gap is on OCR-related tasks.

type of encoder	Model examples
Language supervised	OpenAI CLIP, SigLIP
Self-supervised	DINOv2, I-JEPA
Class supervised	SupViT

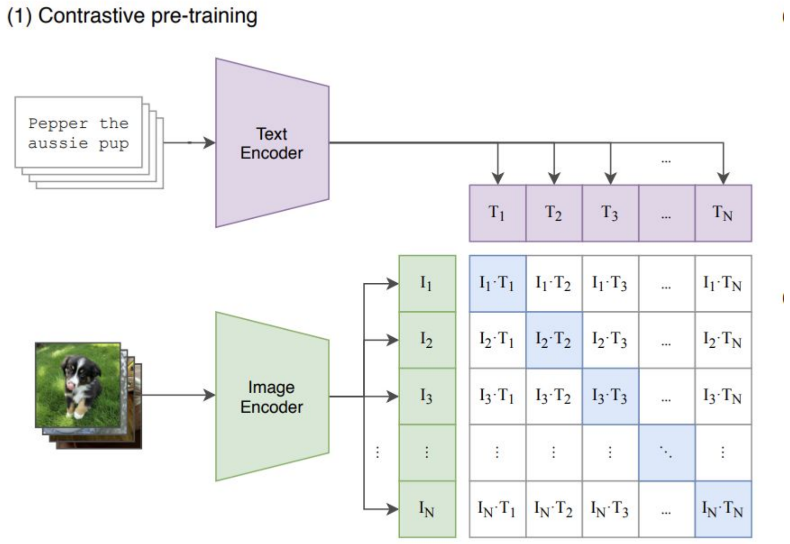
			VQA Benchmarks				MLLM Benchmarks								
	Resolution	**# Tokens**	GQA	Science-QA	TextVQA	AI2D	POPE	MME-P	MME-C	MMB-Cn	MMB-En	SEED-All	SEED-V	SEED-I	MMMU
LLaVA-MORE-3.8B (Ours)															
CLIP ViT-L/14 [51]	336^2	576	62.1	71.3	54.0	61.1	85.9	1372.2	281.1	64.2	69.2	63.5	42.3	69.1	38.8
DINOv2 ViT-L/14 [48]	224^2	256	60.9	66.6	41.4	58.2	85.5	1236.6	281.1	53.8	58.9	59.8	40.6	64.8	37.9
DINOv2$_{reg}$ ViT-L/14 [19]	224^2	256	60.4	69.0	41.3	56.4	85.2	1263.2	**288.2**	57.4	51.4	58.7	41.4	63.2	38.6
SigLIP ViT-L/14 [71]	384^2	729	**63.6**	**73.8**	57.6	**62.9**	86.4	1379.0	282.9	66.5	**71.4**	65.7	46.4	70.8	**40.0**
SigLIP2 ViT-L/14 [66]	384^2	729	63.4	71.8	**59.7**	**62.9**	**86.5**	**1406.7**	282.5	**66.8**	69.8	**66.4**	**47.4**	**71.4**	38.8

https://arxiv.org/pdf/2406.16860 https://arxiv.org/pdf/2503.15621

Contrastive visual encoder

DINOv2 trains on pairs of similar images and focuses solely on visual features, it is not good for cases like text-heavy images.

CLIP is trained with language supervision.



https://arxiv.org/pdf/2304.07193 https://arxiv.org/pdf/2103.00020

Sigmoid loss for language image pre-training

Suggestion is to switch from softmax-based approach to binary classification on the dataset of all pair combinations.

$$-\frac{1}{2|\mathcal{B}|}\sum_{i=1}^{|\mathcal{B}|}\left(\overbrace{\log\frac{e^{t\mathbf{x}_i\cdot\mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|}e^{t\mathbf{x}_i\cdot\mathbf{y}_j}}}^{\text{image}\rightarrow\text{text softmax}}+\overbrace{\log\frac{e^{t\mathbf{x}_i\cdot\mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|}e^{t\mathbf{x}_j\cdot\mathbf{y}_i}}}^{\text{text}\rightarrow\text{image softmax}}\right)$$

where $\mathbf{x}_i = \frac{f(I_i)}{\|f(I_i)\|_2}$ and $\mathbf{y}_i = \frac{g(T_i)}{\|g(T_i)\|_2}$. In this paper, we

$$-\frac{1}{|\mathcal{B}|}\sum_{i=1}^{|\mathcal{B}|}\sum_{j=1}^{|\mathcal{B}|}\underbrace{\log\frac{1}{1+e^{z_{ij}(-t\mathbf{x}_i\cdot\mathbf{y}_j+b)}}}_{\mathcal{L}_{ij}}$$

https://arxiv.org/pdf/2303.15343 – Sigmoid Loss for Language Image Pre-Training

Sigmoid loss for language image pre-training

Algorithm 1 Sigmoid loss pseudo-implementation.

```
# img_emb       : image model embedding [n, dim]
# txt_emb       : text model embedding [n, dim]
# t_prime, b    : learnable temperature and bias
# n             : mini-batch size

t = exp(t_prime)
zimg = l2_normalize(img_emb)
ztxt = l2_normalize(txt_emb)
logits = dot(zimg, ztxt.T) * t + b
labels = 2 * eye(n) - ones(n) # -1 with diagonal 1
l = -sum(log_sigmoid(labels * logits)) / n
```

$$-\frac{1}{|\mathcal{B}|}\sum_{i=1}^{|\mathcal{B}|}\sum_{j=1}^{|\mathcal{B}|}\underbrace{\log\frac{1}{1+e^{z_{ij}(-t\mathbf{x}_i\cdot\mathbf{y}_j+b)}}}_{\mathcal{L}_{ij}}$$

https://arxiv.org/pdf/2303.15343 – Sigmoid Loss for Language Image Pre-Training

Encoder resolution

Each image is represented by a constant number of tokens (256 in case of Gemma 3).

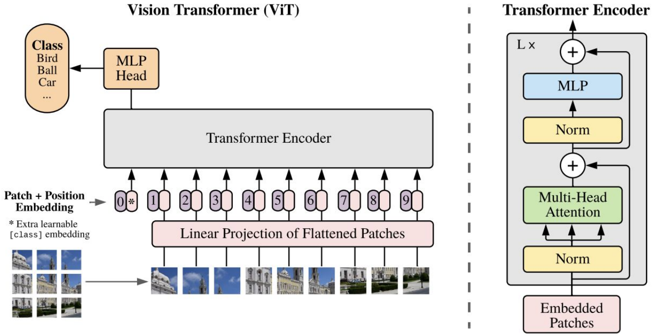


Figure 1: Model overview. We split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder. In order to perform classification, we use the standard approach of adding an extra learnable "classification token" to the sequence. The illustration of the Transformer encoder was inspired by Vaswani et al. (2017).

Resolution	DocVQA	InfoVQA	TextVQA
256	31.9	23.1	44.1
448	45.4	31.6	53.5
896	59.8	33.7	58.0

Table 7 | **Impact of image encoder input resolution.** We measure performance using a short schedule 2B Gemma model on a few evaluation benchmarks to observe the effect of input image resolution on vision encoder pre-training.

Pan & Scan for extreme aspect ratios

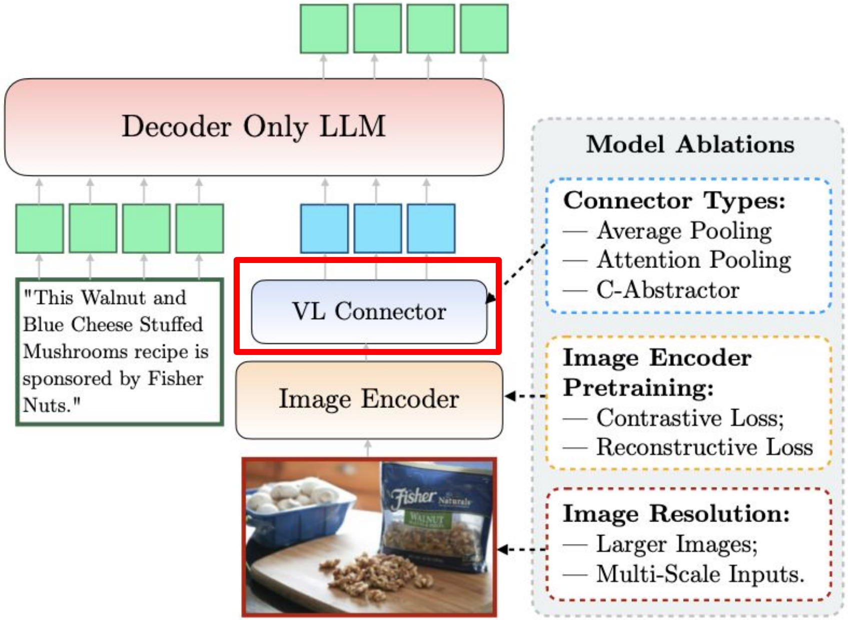
A fixed square resolution results in artifacts when processing non-square **aspect ratios** and high-resolution images. This algorithm segments images into non-overlapping crops of equal size, covering the whole image, and resize them to 896×896 pixels to pass them to the encoder.
Code example link

	DocVQA	InfoVQA	TextVQA
4B	72.8	44.1	58.9
4B w/ P&S	81.0	57.0	60.8
Δ	(+8.2)	(+12.9)	(+1.9)
27B	85.6	59.4	68.6
27B w/ P&S	90.4	76.4	70.2
Δ	(+4.8)	(+17.0)	(+1.6)

Table 8 | **Impact of P&S.** 4-shot evaluation results on the valid set, with and without P&S on a pre-trained checkpoint. Boosts are on tasks associated with images with varying aspect ratios, or involving reading text on images.

https://arxiv.org/pdf/2503.19786 – Gemma 3 Technical Report

VL connectors

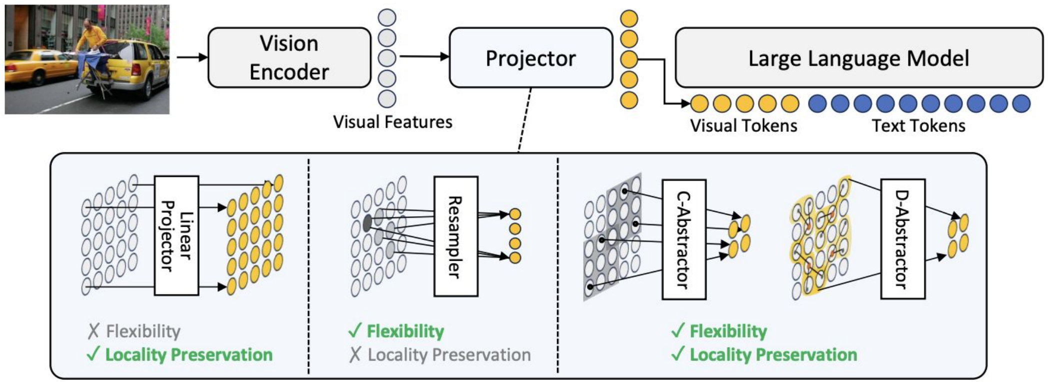


https://arxiv.org/pdf/2403.09611 MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training

VL connectors

Most commonly used are **linear layer** or MLP. We can reduce the number of visual tokens to achieve faster inference times.

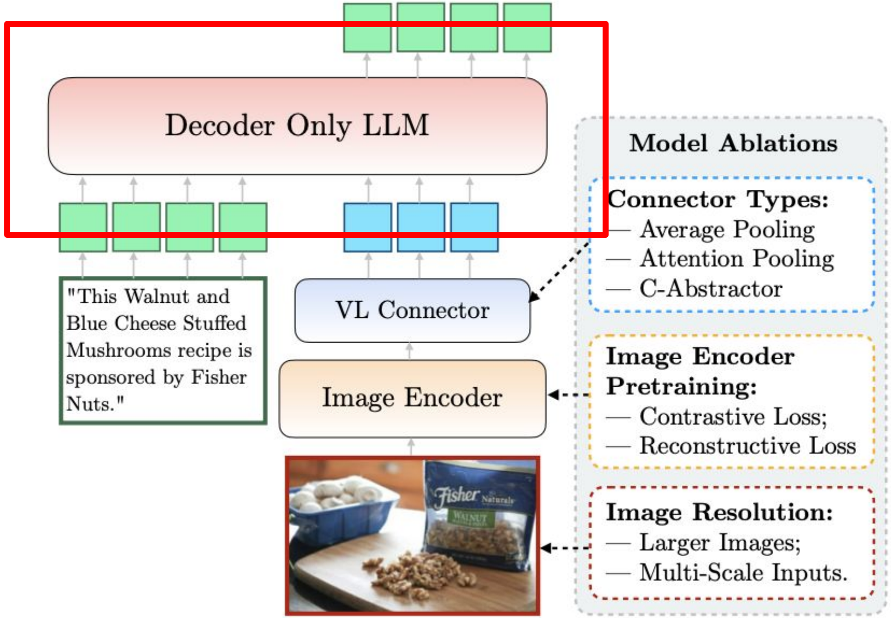
- **average pooling**
- C-Abstractor (convolutions)



(a) Linear Projector (b) Abstractor (c) Locality-enhanced Abstractor

https://arxiv.org/abs/2312.06742 – Locality-enhanced projector for multimodal llm

Textual backbone



https://arxiv.org/pdf/2403.09611 MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training

Textual backbone for VLMs

Main considerations for LLM:

- reasoning capabilities
- long context especially for interleaved
- world knowledge

https://arxiv.org/pdf/2406.16860 – Cambrian-1: A Fully Open, Vision-Centric Exploration of Multimodal LLMs

Visual Instruction Tuning

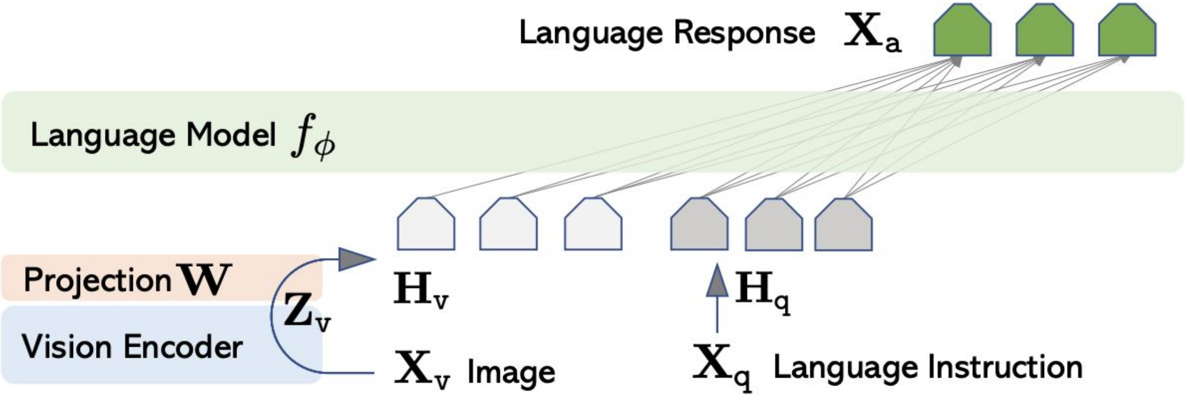


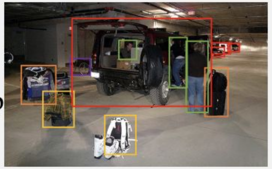
Figure 1: LLaVA network architecture.

Visual Instruction Tuning

Idea is to create vision-language instruction-following data using text-only ChatGPT based on [MS-COCO](#) data. This dataset is 158K unique language-image instruction-following samples in total.

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

https://arxiv.org/pdf/2304.08485 – Visual Instruction Tuning

Visual Instruction Tuning

Training steps:

1. optimize projection **W** from image tokens to LLM embedding space on 500k image captions
2. fine-tuning **W** and **LLM** weights together on 150k dataset where conversations have multiple turns.

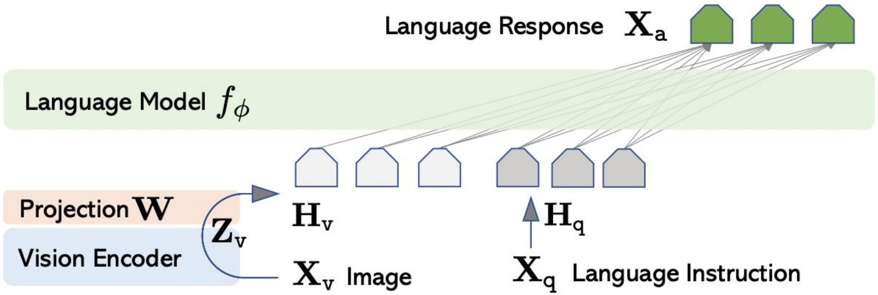


Figure 1: LLaVA network architecture.

Interleaved Multi-Image Instruction Tuning

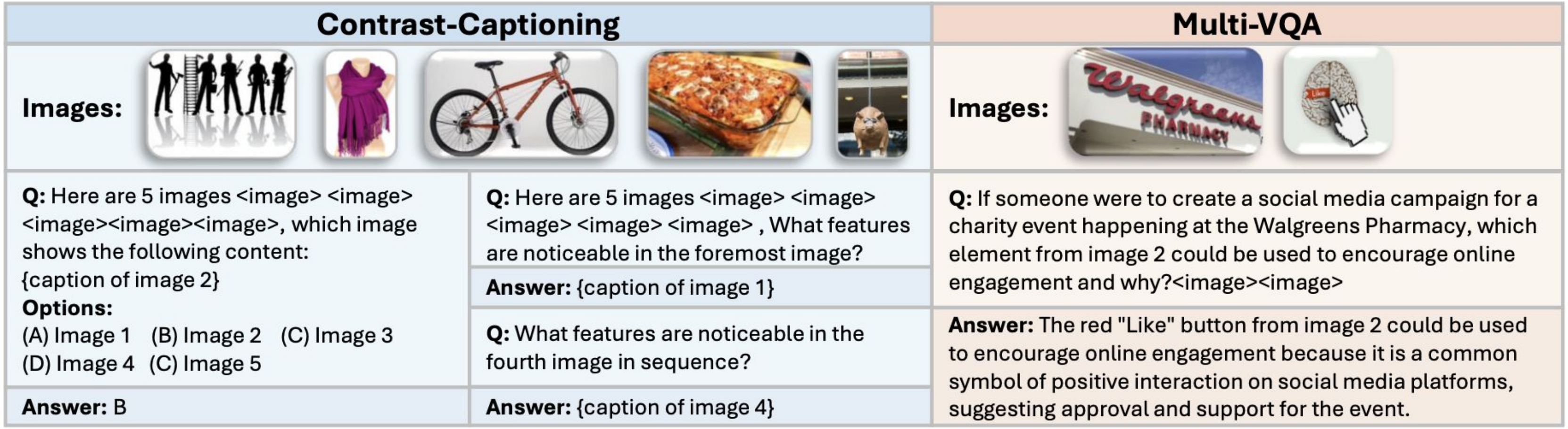
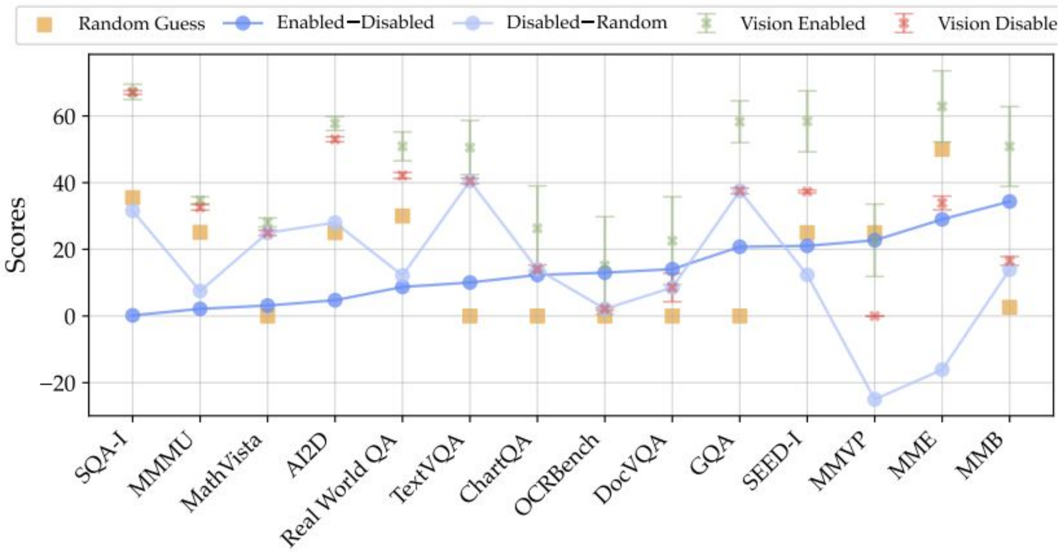


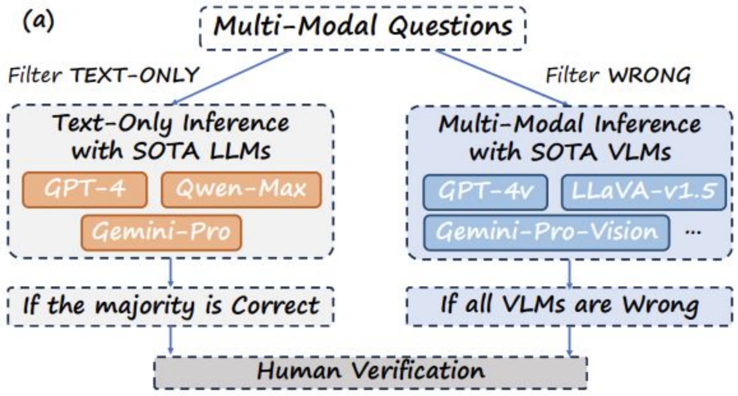
Figure 2: Illustrations of MANTIS-INSTRUCT datasets from Contrast-Captioning and Multi-VQA. Contrastive-Captioning is constructed from existing image captioning datasets to enhance LMM's multi-image 'co-reference' ability. Multi-VQA is GPT-4V synthesized multi-turn multi-image data used to enhance LMM's multi-image 'reason' ability.

https://arxiv.org/pdf/2405.01483 – Interleaved Multi-Image Instruction Tuning

Benchmarking tasks

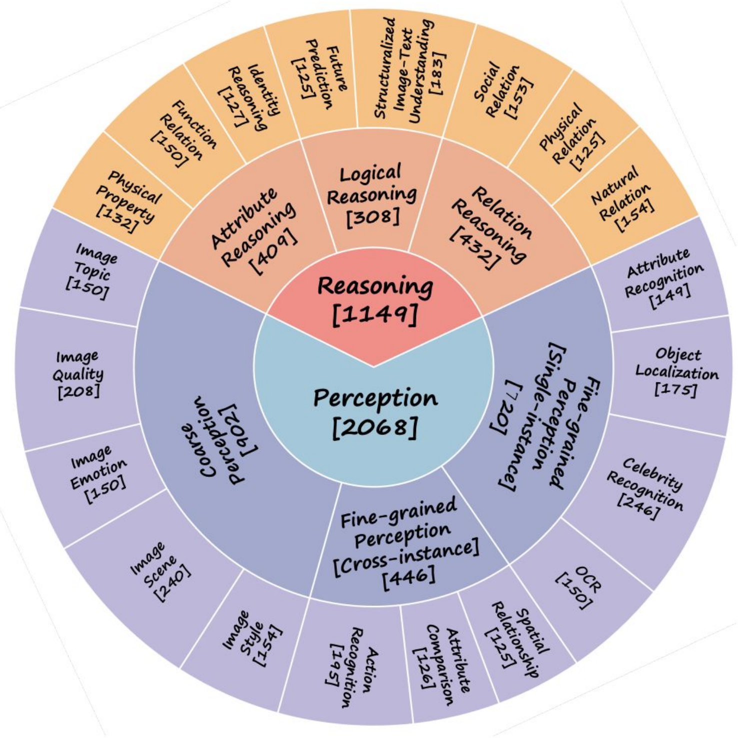
Who's answering the question: the LLM or MLLM?





https://arxiv.org/pdf/2406.16860 – Cambrian-1: A Fully Open, Vision-Centric Exploration of Multimodal LLMs

Benchmarking tasks



https://arxiv.org/pdf/2307.06281 – MMBench

Benchmarking tasks

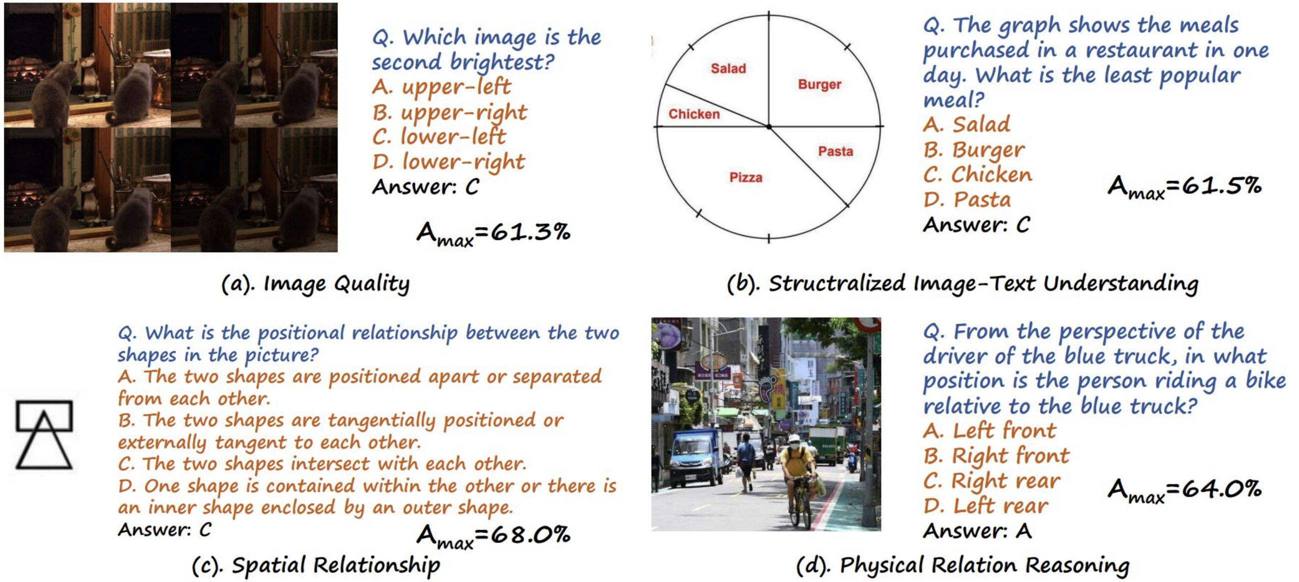


Figure 10: **Hard examples that belong to the 4 L-3 abilities with lowest A_{max}.** All VLMs have made the wrong prediction for the visualized examples under CircularEval.

https://arxiv.org/pdf/2307.06281 – MMBench

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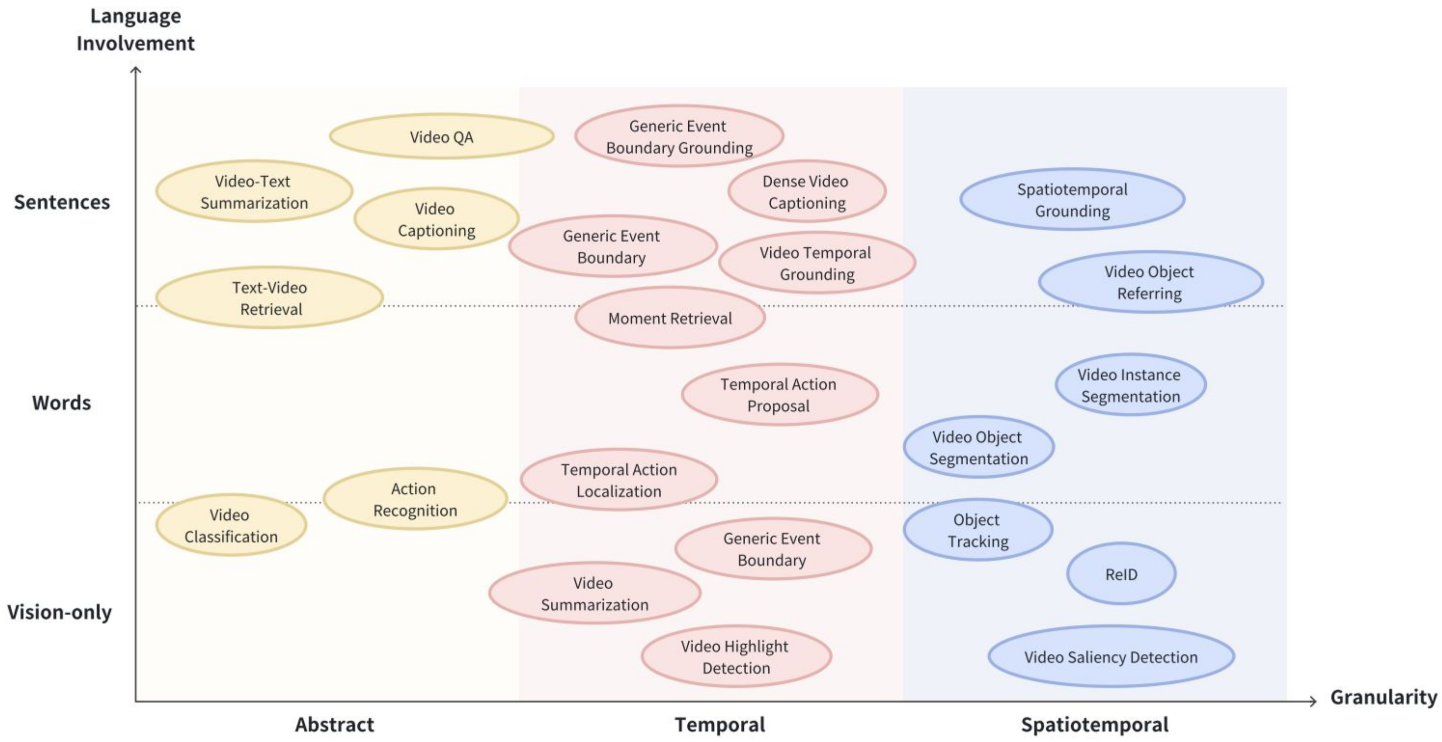
Defining characteristics for VideoLLMs

- Video Encoder – often CLIP ViT-L/14
- Number of seen frames in training – depends a lot on the training data and application
- Adapter – Linear Layer / Q-former
- LLM size (usually 7B/13B/34B)
- Does it utilize audio information? Usually no

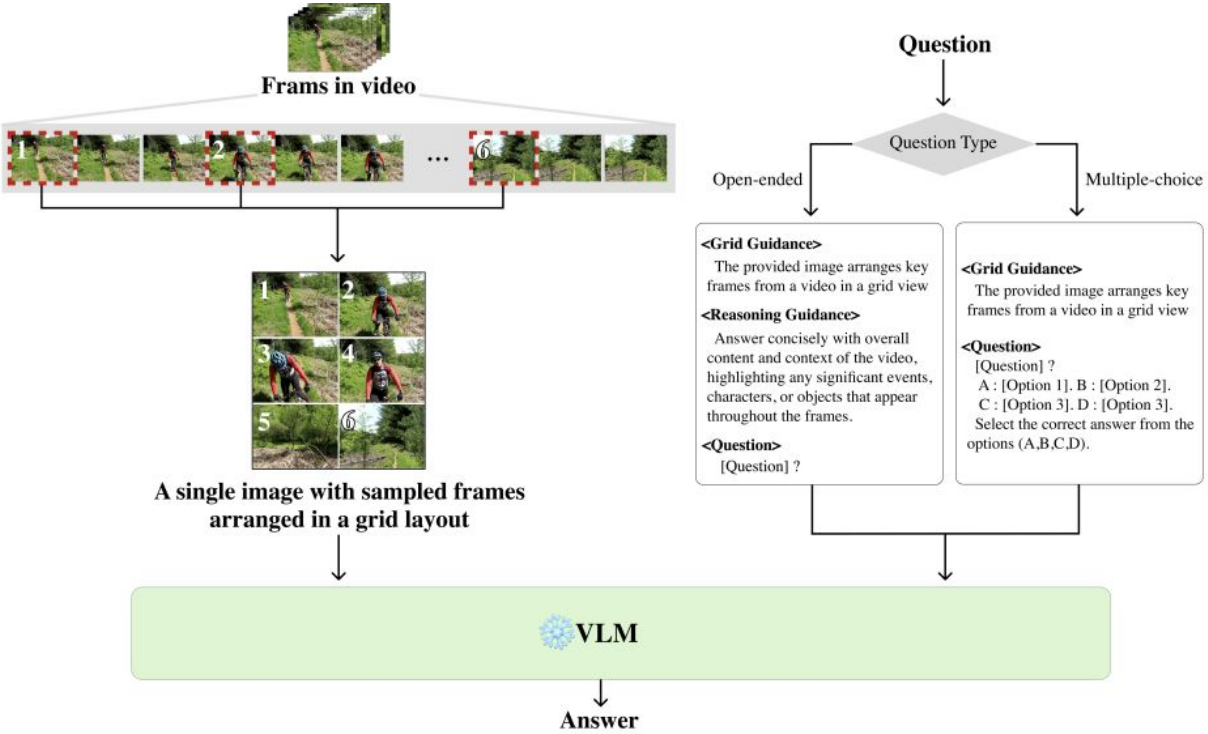
https://arxiv.org/pdf/2408.01319

Benchmarking tasks for video understanding



https://arxiv.org/pdf/2312.17432

An Image Grid Can Be Worth a Video



https://arxiv.org/abs/2403.18406

Temporal and spatial pooling

Pooling on the spatial dimension yields favorable outcomes, whereas temporal dimension pooling is associated with decreased performance for captioning.

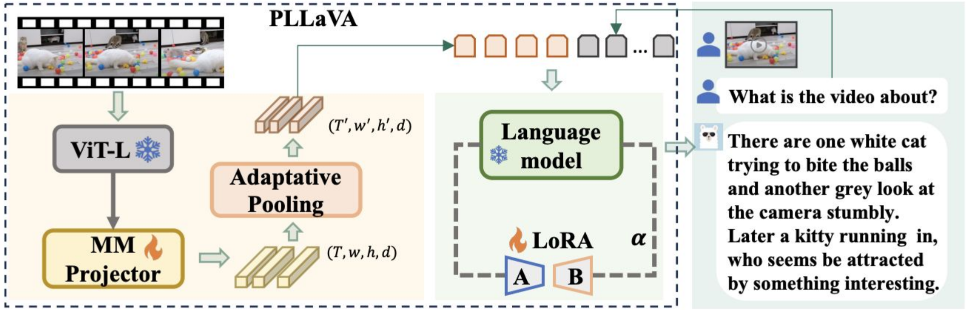


Figure 6: The framework of PLLaVA begins with processing a video from the user through ViT-L and MM projector, yielding visual features with shape (T, w, h, d). These features undergo average pooling, which effectively reduces both temporal and spatial dimensions. The pooled features are then flattened and concatenated with question embeddings, serving as input to the image Large Language Model to generate response to the user. The weights of the image LLMs are fused with LoRA weight learned under video samples.

https://arxiv.org/pdf/2404.16994 – Parameter-free LLaVA Extension

Plan

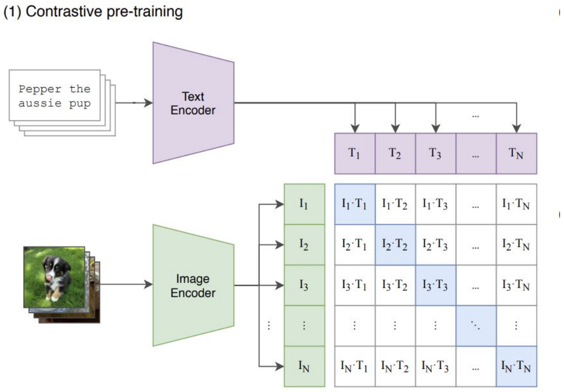
- aligning modalities – contrastive learning with language supervision
  - pre-training
- Visual Language Models
  - visual encoder
  - VL connectors
  - textual backbone
  - instruction tuning
  - evaluation
- Video Understanding
  - training-free adaptation of VLMs to video

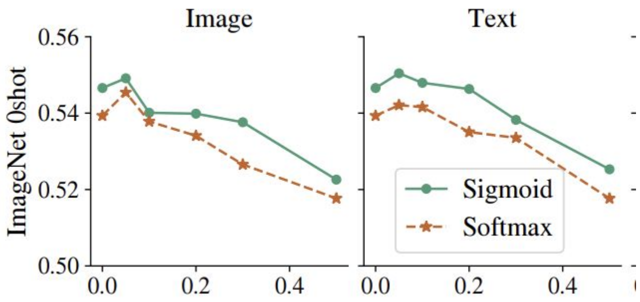
Thanks for attention

Sigmoid loss for language image pre-training

Benefits of SigLIP compared to CLIP:

- can be easily parallelised
- more stable to data noise





https://arxiv.org/pdf/2303.15343 – Sigmoid Loss for Language Image Pre-Training

VL connectors

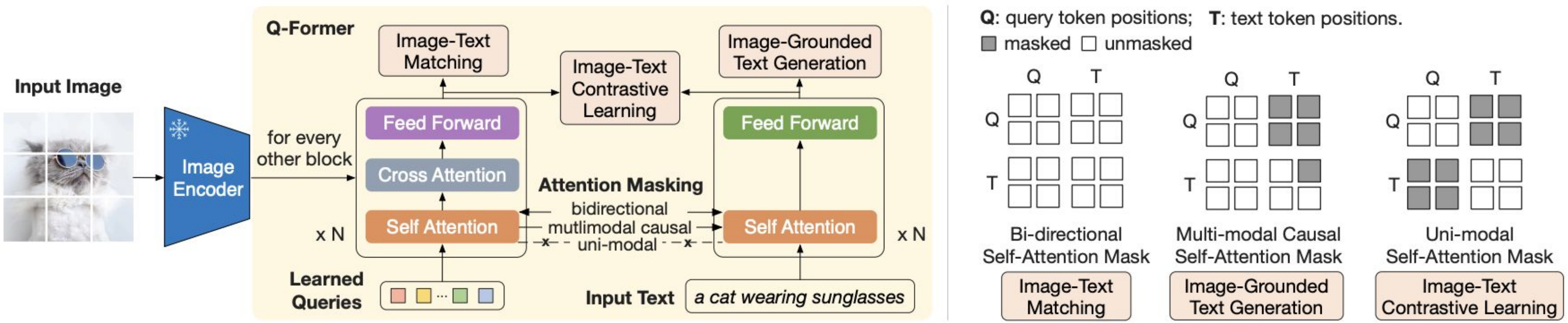


Figure 2. (**Left**) Model architecture of Q-Former and BLIP-2's first-stage vision-language representation learning objectives. We jointly optimize three objectives which enforce the queries (a set of learnable embeddings) to extract visual representation most relevant to the text. (**Right**) The self-attention masking strategy for each objective to control query-text interaction.

https://arxiv.org/pdf/2301.12597 – BLIP-2: Bootstrapping Language-Image Pre-training